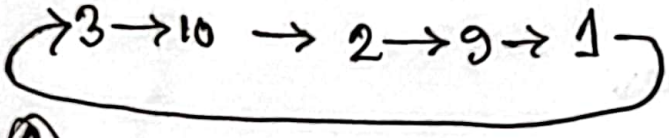
Different types of Linked list:

19-06-23

1) Singly Linked list
2) Singly Circular Linked list →
3) Doubly Linked list
4) Doubly Circular linked list
5) Dummy headed Doubly Linked list
6) Dummy headed Doubly Circular linked list.

1) 3 → 10 → 2 → 9 → 1 → None

2) 

~~for~~ ① Singly Circular Linked list:

```
def forward (head):
    temp_head
    while temp.next != head:
        print (temp. elem)
        temp = temp. next
    print (temp. elem)
```

temp	temp.next	print
3	10	3
10	2	10
2	9	2
9	1	9
1	3	

3 → head

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2) Singly Linked list:

```
def lastnode (head):
    temp = head
    while temp.next != None:
        temp = temp.next
    print (temp.elem)
```

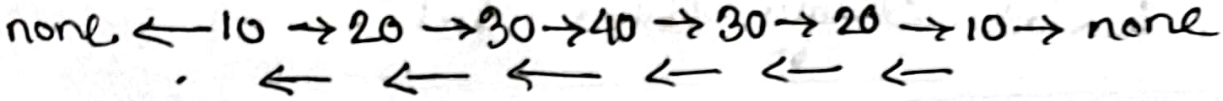
3) Doubly linked list:

None ← 3 → 10 → 2 → 9 → 1 → None

* Node Class:

i) elem ii) next iii) prev

```
def backward (head):
    temp = lastnode (head)
    while temp != None
        print (temp. elem)
        temp = temp. prev.
```



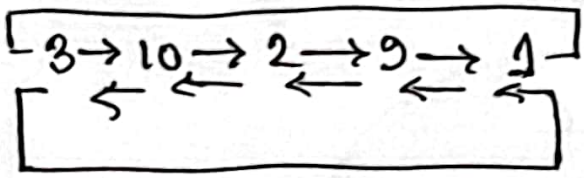
↳ check if palindrome:

new_node.prev = previous_node

(D.I.Y)

④

Doubly Circular Linked list:




```
def forward (head):
    temp = head
    while temp.next != head:
        print (temp.elem)
        temp = temp.next
    print (temp.elem)

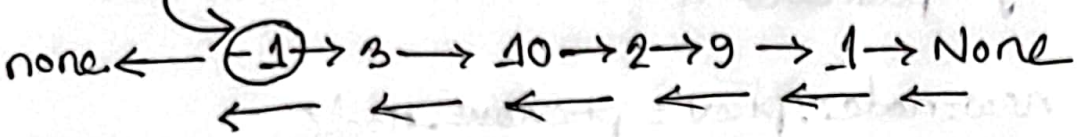
def backward (head):
    temp = head.prev
    while temp != head:
        print (temp.elem)
        temp = temp.prev
    print (temp.elem)
```

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6) Dummy headed Singly linked list:

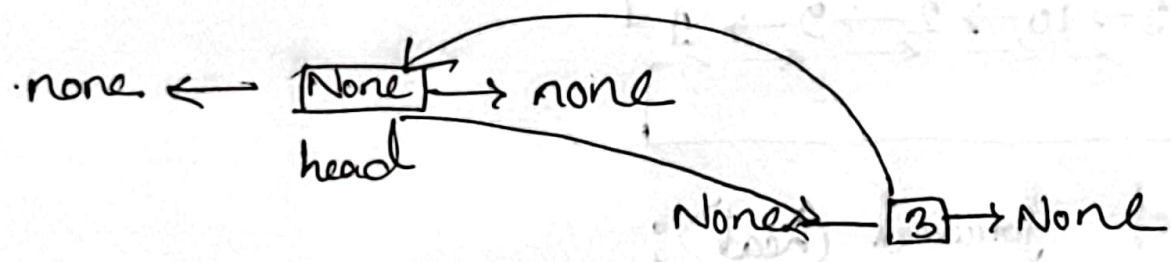
False head node



Creating false head

head = Node (None, None, None)

elem, prev/next, prev/next



n1 = Node (3, None, None)

head.next = n1

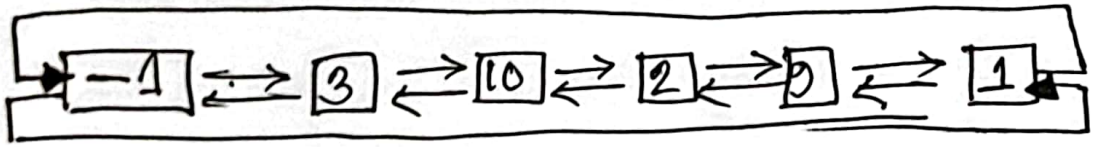
n1.prev = head

```
def forward (head):
    temp = head

def backward (head):
    temp = lastNode (head)
    while temp != head:
        temp = temp.prev.
```

(6)

Dummy Headed Doubly Circular Linked List:




```
def forward (head):
    temp = head.next
    while temp != head:
        .
        .
        .
        temp = temp.next
```

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```
def backward (head):
    temp = head.prev
    while temp != head
        temp = temp.prev.
```